



Check out the Robots in Love album in the DVD by Berteycox which explores the world from the eyes of a robot

Kevin Warwick

The Cybernetics professor who calls himself the world's first cyborg, with computer chips implanted in his left arm

as they had to compete with other (hostile) communities. Present industrial systems are more complex and multiple mobile robots are increasingly preferred. Multiple mobile robots are not spatially constrained and performance benefits are manifold. Robots/agents equipped with knowledge, motivations, reasoning and planning capabilities may communicate each other and, share data and information. Cooperation protocol by distributed control, effective communication and fault tolerance while having efficiency of cooperation, adaptation and robustness are some of the research directions associated with multi-robot / agent systems.

Robot soccer

The robot soccer platform serves as a test bed to study and research on the issues pertaining to cooperative mobile robots. A robot team needs to coordinate its actions while competing with another team. The robot soccer system also has an explicit performance measure, the match score. Cooperation, decision making, planning, modelling, learning, robot architecture, vision tracking algorithm, sensing and communication are some of the directions of study. Organisations such as FIRA (www.fira.net) are promoting robot soccer as a competitive platform to push technology. In robot soccer, teams have to coordinate and compete while chasing an indivisible resource (the ball). Robot soccer, therefore, is a benchmark problem to study various issues along coordination / cooperation and competition, giving insights into problems in social / life sciences. International robotic competitions

have played an important role in pushing technologies leading to a number of technology start ups. The FIRA Robot World Cup and Congress Bangalore 2010 (www.fira.

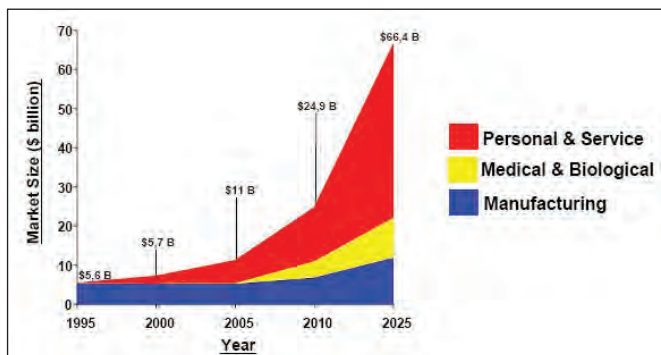
in) is expected to act as a catalyst in these directions where the advancements in robotics will be showcased. FIRA India aims to propagate and popularise robotics and international-level robotic competitions across India. The objective of the competition is to promote the spirit of science and technology among the younger generation. It serves to bring together skilled researchers and students from different backgrounds to challenge one another in developing better robots for technologies of the future.

Robotic soccer competitions have emerged as benchmark problems for the integration of several technologies required to build sophisticated autonomous robots. These competitions are meant to stimulate research and serve as a platform for the exchange of ideas for the researchers involved. We've set 2050 as our target to pitch a team of humanoids against humans to play the game of football. Although many of us are sceptical of this deadline, primarily because technology is yet to reach the needed threshold for this to materialise. What these deadlines do, is help us work towards its accomplishment and push ourselves.

The path ahead

We expect robots will be utilised in the 21st century for household applications. This will pave the way for advanced robotic technologies to dominate in the 21st century. Some day, personal robots will be as popular as personal computers!

There are opportunities in industrial robotics space for further development specific to Indian requirements and expansions along manufacturing. Mobile robotics is relatively new, and has higher



Due to the restricted growth in recent times, robotics, as a sector is expected to witness an exponential growth rate

potential growth in the coming decades, where a variety of applications right from consumer products to medical robotics are expected to usher in avenues.

Higher education in India is often neglected and as a result, there's an acute shortage of skilled man power in specialised areas, especially in the robotics and intelligent systems industry. The number of institutions imparting courses related to robotics design is limited in India. According to a recent NASSCOM study, 75 per cent of Indian engineers are not employable and IT firms reject about 90 per cent of Indian graduates. The employability of graduates is directly related to the quality of education, availability of state-of-the-art facilities and the assessment methodologies adopted. Timely reforms in the educational sector can assist India to ride the robotics wave and to position better. Japanese and Korean domestic robotic markets are able to sustain their growths from within. Governmental support and push along niche areas of research in interactive and intelligent robotic systems can help India to position itself better and to tap into the immense potential that lay ahead. **d**



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